

# ANDREW I.L. WILLIAMS

Scripps Institution of Oceanography, UC San Diego

**Email:** andrewwilliams@ucsd.edu // **Citizenship:** Dual U.K. and U.S.A.

## EDUCATION

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**University of Oxford**  
DPhil (PhD) in Climate Physics

*October 2019 - July 2023*  
Advisor: Philip Stier

**University of Oxford**  
Bachelors & Masters degrees in Physics

*October 2015 - June 2019*  
Classification: 1<sup>st</sup>

## ACADEMIC APPOINTMENTS

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**Postdoctoral Scholar**  
Scripps Institution of Oceanography

October 2025 - present

**Postdoctoral Research Associate**  
Princeton University & NOAA GFDL

August 2023 - August 2025

**Visiting Researcher**  
TU Delft

June 2022 - July 2022

**Visiting Researcher**  
Max Planck Institute for Meteorology

April 2022 - May 2022

**Research Associate**  
Massachusetts Institute of Technology

June 2019 - January 2020

**Summer Undergraduate Research Fellow**  
California Institute of Technology

Summer 2018

## AWARDS

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**Outstanding Early Career Presentation Award**  
CFMIP-GASS meeting, Paris

2023

**Outstanding Student and PhD candidate Presentation Award**  
EGU

2022

**Outstanding Student Presentation Award**  
AGU Fall Meeting

2022

**Moritz-Heyman Scholarship**  
Scholarship for low-income students who won a place at Oxford University (£16,000 total).

2015-2019

**St. Hilda's College, 125th Anniversary Scholarship**  
Prize for high grades in first year examinations at Oxford (£1,250 total)

2019

## GRANTS & FUNDING

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**Scripps Institutional Postdoctoral Fellowship**  
Scripps Institution of Oceanography, UCSD ( $\approx$  \$200,000)

2026-2028

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| <b>CIMES Postdoctoral Fellowship</b><br>Princeton University ( $\approx$ \$200,000)                                                            | 2023-2025 |
| <b>NOAA Climate &amp; Global Change Postdoctoral Fellowship</b><br>Yale University (declined) ( $\approx$ \$200,000)                           | 2023-2025 |
| <b>NERC PhD Studentship</b><br>Fully funded doctoral fellowship at the University of Oxford. ( $\approx$ £100,000)                             | 2019-2023 |
| <b>Laidlaw Research and Leadership Scholarship</b><br>Awarded to fund research at MIT with Prof. Paul O’Gorman. ( $\approx$ £10,000)           | 2019      |
| <b>Caltech Summer Undergraduate Research Fellowship</b><br>Awarded to fund research at Caltech with Prof. Tapio Schneider. ( $\approx$ £8,000) | 2018      |

## PUBLICATIONS

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### Manuscripts in preparation

- Risi, C. and co-authors including Williams, A. I. L.: Amplification of temperature changes with altitude in the tropics and subtropics  
*Reviews of Geophysics*
- Watson-Parris, D. and co-authors including Williams, A. I. L., ClimateBench v2.0: Probabilistic Climate Model Scoring
- Williams, A. I. L.: Tropical circulation as the statistics of convection

### 2026

- C. Proistosescu, P. Paul, N.J. Lutsko, Williams, A. I. L. & M.F. Stuecker, The forgotten role of wave dynamics in modulating the low cloud response to warm pool warming  
*Geophysical Research Letters* (submitted)  
[arxiv.org/abs/2603.24488](https://arxiv.org/abs/2603.24488)
- Davenport, D., Madan, V. and co-authors including Williams, A. I. L., JCM v1.0: A Differentiable, Intermediate-Complexity Atmospheric Model  
*Geoscientific Model Development* (submitted)  
[10.5194/egusphere-2025-6266](https://doi.org/10.5194/egusphere-2025-6266)
- Williams, A. I. L., Bridging Clarity and Accuracy: A Simple Spectral Longwave Radiation Scheme for Idealized Climate Modeling  
*Journal of Advances in Modeling Earth System*  
[10.1029/2025MS005405](https://doi.org/10.1029/2025MS005405)
- Williams, A. I. L. & Merlis, T.M., State-dependence of polar amplification in an idealized GCM  
*Geophysical Research Letters*  
[10.1029/2025GL118423](https://doi.org/10.1029/2025GL118423)

### 2025

- Williams, A. I. L. & Jeevanjee, N., Sea-surface temperature patterns, radiative cooling, and hydrological sensitivity  
*Geophysical Research Letters*  
[10.1029/2025GL117734](https://doi.org/10.1029/2025GL117734)
- Williams, A. I. L. & Jeevanjee, N., A robust constraint on the response of convective mass fluxes to warming  
*Journal of Advances in Modeling Earth Systems*  
[10.1029/2024MS004695](https://doi.org/10.1029/2024MS004695)

- Watson-Parris, D. and co-authors including Williams, A. I. L., Weak surface temperature effects of recent reductions in shipping SO<sub>2</sub> emissions, with quantification confounded by internal variability  
*Atmospheric Chemistry & Physics*  
[10.5194/egusphere-2024-1946](#)
- Mackie, A. , Byrne, M.P., van de Koot, E. & Williams, A. I. L., Circulation and cloud responses to patterned SST warming  
*Geophysical Research Letters*  
[10.22541/essoar.172736799.93094846](#)
- Herbert, R. J., Williams, A. I. L., Weiss, P., Klocke, D. & Stier, P., Isolating aerosol-climate interactions in global storm-resolving simulations  
*Atmospheric Chemistry & Physics*  
[10.5194/egusphere-2024-1689](#)

## 2024

- Schmidt, H. & co-authors including Williams, A. I. L., Effects of vertical grid spacing on the climate simulated in the ICON-Sapphire global storm-resolving model  
*Geoscientific Model Development*  
[10.5194/gmd-17-1563-2024](#)
- Bloch-Johnson, J. and co-authors including Williams, A. I. L., The Green's Function Model Intercomparison Project (GFMIIP) Protocol  
*Journal of Advances in Modeling Earth Systems*  
[10.1029/2023MS003700](#)

## 2023

- Dagan, G., Yeheskel, N. & Williams, A. I. L., Enhanced radiative forcing from aerosol-cloud interactions due to large-scale circulation adjustments  
*Nature Geoscience*  
[10.1038/s41561-023-01319-8](#)
- Williams, A. I. L., Watson-Parris, D., Dagan, G. & Stier, P., Dependence of fast changes in global and local precipitation on the geographical location of absorbing aerosol  
*Journal of Climate*  
[10.22541/au.167364749.93845737/v1](#)
- Williams, A. I. L., Jeevanjee, N. & Bloch-Johnson, J., Circus Tents, Convective Thresholds and the Non-Linear Climate Response to Tropical SSTs  
*Geophysical Research Letters*  
(Editor's Highlight)  
[10.1029/2022GL101499](#)

## 2022

- Williams, A. I. L., Stier, P., Dagan, G. & Watson-Parris, D., Strong control of effective radiative forcing by the spatial pattern of absorbing aerosol  
*Nature Climate Change*  
(Press coverage)  
[10.1038/s41558-022-01415-4](#)
- Dagan, G., Stier, P., Dingley, B. & Williams, A. I. L., Examining the regional co-variability of the atmospheric water and energy imbalances in different model configurations - linking clouds and circulation  
*Journal of Advances in Modeling Earth Systems*  
[10.1029/2021MS002951](#)
- Williams, A. I. L. & O'Gorman, P. A., Summer-Winter Contrast in the Response of Precipitation Extremes to Climate Change over Northern Hemisphere Land  
*Geophysical Research Letters*  
[10.1029/2021GL096531](#)

## 2021

- Watson-Parris, D., Williams, A. I. L., Deaconou, L. & Stier, P., Model calibration using ESEm v1.0.0 - an open, scalable Earth System Emulator  
*Geoscientific Model Development*  
[10.5194/gmd-14-7659-2021](https://doi.org/10.5194/gmd-14-7659-2021)

## PRESENTATIONS

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### 2026

- **EGU General Assembly Meeting (upcoming)** Talk  
Tropical circulation as the statistics of convection
- **3rd Annual Pre-EGU Ice Cloud Workshop, Vienna (upcoming)** Invited talk  
Physics of tropical convection and circulation in a changing climate
- **University of Vienna** Invited talk  
A robust constraint on the response of convective mass fluxes to warming
- **Scripps Institution of Oceanography** Invited talk  
Hierarchical Climate Dynamics: From the Tropics to the Poles
- **University of Chicago** Invited talk  
Hierarchical Climate Dynamics: From the Tropics to the Poles
- **Colorado State University** Invited talk  
Hierarchical Climate Dynamics: From the Tropics to the Poles
- **Lamont-Doherty Earth Observatory** Invited talk  
Hierarchical Climate Dynamics: From the Tropics to the Poles

### 2025

- **Texas A&M University** Invited talk  
Tales of climate dynamics & outreach
- **University of Exeter** Invited talk  
State-dependence of polar amplification in an idealized, ice-free GCM
- **Gordon Research Conference on Radiation and Climate** Poster  
A Simple Spectral Longwave Radiation Scheme for Idealized Climate Modeling
- **Florida State University** Invited talk  
The physics of tropical convection and the large-scale circulation in a warming climate
- **ECS & Cloud Feedbacks Virtual Symposium** Talk  
How do convective mass fluxes respond to global warming?
- **EGU General Assembly Meeting** Talk  
A robust constraint on the response of convective mass fluxes to warming
- **University of Washington** Invited talk  
A robust constraint on the response of convective mass fluxes to warming

### 2024

- **GFDL Lunchtime Seminar**  
A robust constraint on the response of convective mass fluxes to warming
- **CLIVAR TROPICS working group, Hamburg** Invited talk  
Is there a “hydrological constraint” on tropical circulation weakening?
- **City College of New York** Invited talk  
Physical constraints on the response of convective mass fluxes to warming
- **AOFD** Talk  
A robust constraint on the response of convective mass fluxes to warming

- **CFMIP** Talk  
A robust constraint on the response of convective mass fluxes to warming
- **Hebrew University of Jerusalem** Invited talk  
A robust constraint on the response of convective mass fluxes to warming
- **Princeton University** Talk  
A robust constraint on the response of convective mass fluxes to warming
- **Columbia University** Invited talk  
A robust constraint on the response of convective mass fluxes to warming

## 2023

- **University of East Anglia** Invited talk  
Non-linear climate response to tropical SSTs explained by a convective threshold
- **Center for Atmosphere Ocean Science, NYU** Invited talk  
Non-linear climate response to tropical SSTs explained by a convective threshold
- **Reading University** Invited talk  
Do convective mass fluxes constrain the tropical circulation response to warming?
- **Gordon Research Conference on Radiation and Climate** Poster  
Circus tents, convective thresholds and the non-linear climate response to tropical SSTs
- **CFMIP-GASS meeting** Talk  
Circus tents, convective thresholds and the non-linear climate response to tropical SSTs  
(Winner of an Outstanding Early Career Presentation Award)
- **ECS & Cloud Feedbacks Virtual Symposium** Talk  
Circus tents, convective thresholds and the non-linear climate response to tropical SSTs
- **4th biennial workshop on the regional climate response to aerosol** Talk  
Understanding the dependence of fast changes in global and local precipitation on the geographical location of absorbing aerosol

## 2022

- **AGU Fall Meeting** Talk  
Circus tents, convective thresholds, and the non-linear climate response to tropical SSTs
- **NOAA GFDL** Invited talk  
Strong control of effective radiative forcing by the spatial pattern of absorbing aerosol
- **Princeton University** Invited talk  
Circus tents, convective thresholds, and the non-linear climate response to tropical SSTs
- **Yale University** Invited talk  
Non-linearities in the pattern effect explained by a convective threshold
- **3rd Pan-GASS Meeting, Monterey** Poster  
Impact of warm-rain suppression on the climate of a mock-Walker circulation
- **TU Delft** Invited talk  
Clouds, aerosols and the global circulation
- **2nd Workshop on Cloud Organization, Utrecht** Poster  
Aerosol-cloud-circulation interactions in cloud-resolving simulations with an imposed SST gradient
- **CLIVAR Pattern Effect Workshop** Poster  
SST Green's Functions for regional precipitation
- **EGU General Assembly Meeting** Invited talk  
Strong control of effective radiative forcing and precipitation by the spatial pattern of absorbing aerosol  
(Winner of an Outstanding Student Presentation Award)

## 2021

- **AGU Fall Meeting** Talk  
 Contrasting Seasonal Response of Northern Hemisphere Precipitation Extremes to Climate Change  
 (Winner of an Outstanding Student Presentation Award)
- **AGU Fall Meeting** Poster  
 Understanding the “pattern effect” of absorbing aerosol

## 2019

- **EGU General Assembly Meeting** Poster  
 Optimizing the number of convective plumes in EDMF cloud parameterization schemes using high-resolution LES simulations

## TEACHING

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- Mentoring of a GFDL summer intern** 2024-present  
*Training and mentoring of a student from University of Puerto Rico at Mayagüez, working on the response of rainfall extremes to CO<sub>2</sub>. I have continued meeting with the student to support them as they progress through graduate school.*
- Guest Lecturer, Colorado State University** 2024  
*Gave a guest lecture on non-linear aspects of Earth’s energy balance*
- Guest Lecturer, City University of New York** 2023  
*Gave a guest lecture on basic aspects of Earth’s atmospheric circulation*
- Supervision of an Oxford Masters’ student** 2022-2023  
*Day-to-day supervision of a Master’s student studying the response of regional precipitation to SST anomalies*
- Teaching Assistant: Atmospheric Physics Master’s Course** 2020-2022  
*Marked homework assignments and provided weekly feedback on students’ work.*
- Lecturer / Course Lead** 2020-2023  
*Designed and delivered a course for first-year PhD students on modeling the global-mean climate using energy balance models. Delivered blackboard lectures and interactive Python coding demonstrations.*

## DIVERSITY & OUTREACH EFFORTS

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- The Weather & Climate Livestream ([wclivestream.com](https://wclivestream.com))** 2025 – present  
 Co-organizer of continuous, multi-day livestream events featuring talks by US-based weather and climate scientists, advocating for continued federal investment in weather and climate research. The inaugural 2025 event reached hundreds of thousands of viewers, generated over 30,000 calls to Congressional representatives, and was covered by the New York Times.
- Climate Up Close ([climateupclose.org](https://climateupclose.org))** 2024 – present  
 Small team of climate scientists committed to making the essentials of climate science accessible to a broad audience through public lectures, Q&A sessions and demos.
- Harlem StreetSquash volunteer** 2023 – 2024  
 Providing weekly math and physics tuition to students and young adults in Harlem, with the aim to ensure all participants graduate from high school, enroll in and complete a post-secondary program, and gain meaningful employment. I also assist in running weekly squash training sessions for the students.
- GFDL Diversity, Equity, Inclusion & Accessibility Committee** 2023 – 2025  
 Development and execution of DEIA efforts to monitor and enhance lab culture and community at GFDL. Activities included: outreach efforts with minority serving institutions, organizing community events to promote wellness and community within the lab and promoting diversity in hiring practices at all levels of the lab.

**AOPP Equality, Diversity & Inclusion Committee**

2020 – 2023

Organized the department's first student-led EDI group, which aims to develop and sustain a diverse, inclusive, and equitable academic environment and community. Activities included: collecting demographic data on student applicants to inform access efforts, setting up a community EDI library and arranging accessible coffee breaks and social events for the department.

**PhD mentoring scheme**

2021 – 2023

Developed a mentoring scheme which matches incoming PhD students with a postdoc or more senior PhD student who can provide advice on adjusting to graduate study at Oxford. Scheme was trialed at a small-scale before being rolled out across the entire Oxford Physics department.

**Oxford Sparks**

2020 – 2021

Recorded an [outreach video about clouds](#) which has reached over 150,000 people across social media.

**Seren Hub**

2016 – 2023

Provided interview practice and entrance exam help for Welsh students from disadvantaged backgrounds who want to study Physics or Mathematics at university.

## ACADEMIC SERVICE

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**Peer reviewer** for *Nature*, *Nature Geoscience*, *Nature Climate Change*, *npj Climate and Atmospheric Science*, *Journal of Climate*, *Journal of Advances in Modeling Earth Systems*, *Scientific Reports*, *Atmospheric Chemistry and Physics*, *Geophysical Research Letters*, *Geoscientific Model Development*.

**Proposal Reviewer** for *NSF-AGS* and *Department of Energy*.

**Session Convener** 2025-present  
*“Climate Sensitivity, Radiative Feedbacks and the Pattern Effect” at the EGU General Assembly Meeting*

**Organizing Committee - CFMIP 2024** 2024  
*CFMIP conference 2024, Boston College // [website link](#)*

**Organizing Committee** 2023-present  
*GFDL’s Climate Sensitivity Journal Club*

**Steering Committee** 2023-present  
*Climate Sensitivity & Cloud Feedback Virtual Symposium // [website link](#)*

**Session chair** 2022  
*Chair of session on “Absorbing Aerosols: Experiments, Observations, and Modelling” at the EGU General Assembly Meeting 2022*

**Policy briefing** Nov-Dec 2020  
*Commissioned by Shadow Secretary for Health and Social Care to research the intersection between ‘Pandemics and Climate Change’. Findings were written in a [white paper](#) and presented to the Government.*